DIT UNIVERSITY
Dehradun

School of Computing

Course Description Document

of

Deep Learning (CSN342)

3 Year (6th Semester)

Course Structure & Syllabus of B.Tech.–Computer Science & Engg.
Applicable for Batch: 2022-26

Course Coordinator:

Name	Designation	Email	Cabin
Dr. Anushikha Singh	Assistant Professor	Anushikha.singh@dituniversity.edu.in	VED210

S. No	Faculty Name	Cabin No	Mail_Id	Theory/Lab	Section
1.	Dr. Anushikha Singh	VED210	anushikha.singh@dituniversity.edu.in	Theory, Lab	A, B, C and D (Theory & Labs) and E (Lab)
2.	Dr. Vyomika Singh	Available ONLY on DIT email	vyomika.singh@dituniversity.edu.in	Theory, Lab	E, F, G and H (Theory) and G, H, I J and F (Labs)
3.	Dr. Gaurav Dhiman	VED 419	gaurav.dhiman@dituniversity.edu.in	Theory	I and J (Theory)

Course Summary

1. School offering the course	School Of Computing
2. Course Code	CSN342 (DL)
3. Course Title	**Deep Learning**
4. Credits (L:T:P:C)	2:0:1:3
5. Contact Hours (L:T:P)	2:0:2
6. Prerequisites (if any)	Machine Learning (CSN344)
7. Course Basket	Discipline Elective

Course Summary:

This course aims to provide a basic understanding of deep learning concepts, implementation of supervised and unsupervised algorithms. This course introduces students to enterprise data and the process and technologies to integrate data from a variety of sources.

Course Objectives

The objective of this course is to cover the fundamentals of neural networks as well as some advanced topics such as recurrent neural networks, long short-term memory cells and convolutional neural networks. The course also requires students to implement programming assignments related to these topics.

Course Outcomes

- Understand the concept of artificial neural networks, convolutional neural networks, and recurrent neural networks
- Discuss how to speed up neural networks along with regularization techniques to reduce overfitting
- Understand the concept of generative models
- Implement deep learning algorithms, and learn how to train deep network

Curriculum Content

UNIT 1: Fundamentals of neural networks [5L]

Biological Neuron vs. Artificial neuron, Idea of computational units, McCulloch–Pitts unit and Thresholding logic, Linear Perceptron, Perceptron Learning Algorithm, Linear separability. Convergence theorem for Perceptron Learning Algorithm, Implementation of Boolean functions using Perceptron.

UNIT 2: Feed Forward Networks [5L]

Multilayer Perceptron (MLP) and activation functions. Logistic Regression, Learning parameters, Loss functions, Empirical Risk Minimization. Gradient Descent, Backpropagation algorithm. Regularization techniques. Difficulty of training deep neural networks, Greedy layer wise training.

UNIT 3: Advanced training methods for Deep Neural Networks and Autoencoders [5L]

Better Training of Neural Networks: Advanced optimization methods for neural networks (Adagrad, adadelta, Rmsprop, adam, NAG), second order optimization methods for training. Critical points, Saddle point problem in neural networks. Introduction to Autoencoders: Structure and Functionality.

UNIT 4: Convolution neural networks and recurrent neural networks [5L]

Convolutional Neural Networks: The convolution operations: Filter/kernel, Stride, Padding, Pooling, CNN Architectures: LeNet, AlexNet. Transfer Learning and Pre-trained CNNs. Sequence learning problems, Recurrent Neural Networks, Back propagation through time, Long Short-Term Memory, Gated Recurrent Units, Bidirectional LSTMs, Bidirectional GRUs.

UNIT 5: Generative AI and Application of Deep Learning [5L]

Markov Networks, Restricted Boltzmann Machines, Gibb's sampling. Generative Adversarial Networks: The Generator and Discriminator Architecture, Loss functions and Training strategies. Applications of Deep Learning: Computer Vision, Natural language processing, Speech processing and Deep Learning Tools.

TEXT BOOKS:
1. Deep Learning, Ian Goodfellow and Yoshua Bengio and Aaron Courville, MIT Press, 2016.

REFERENCES:
1. Raul Rojas, Neural Networks: A Systematic Introduction, Springer-Verlag, Berlin, New-York, 1996
2. Christopher Bishop, Pattern Recognition and Machine Learning, 2010
2. Pattern Recognition and Machine Learning, Christopher Bishop, 2007

TEACHING AND LEARNING STRATEGY

All materials (ppts, assignments, labs, etc.) will be uploaded on LMS. Refer to your course in LMS Teams for details.

LIST OF EXPERIMENTS:

S. No.	Title of experiment
1.	Exp 1: Perceptron and Multilayer Perceptron (MLP) a. Implementation of Boolean functions using McCulloch-Pitts (MP) neuron b. Implementation of Boolean functions using Perceptron and MLP
2.	Exp 2: Optimization of Neural Network a. Gradient descent optimization techniques (SGD, Momentum, RMSprop, Adam)

	b. Implementation of Back propagation algorithm
3.	Exp 3: Better training of neural network a. Greedy Layer wise Training of a feed-forward neural network b. Techniques to improve training performance (Batch Normalization, Dropout, Data Augmentation) c. Performance evaluation (Loss curves, Accuracy, Confusion Matrix)
4.	Exp 4: Implementation of Convolutional Neural Networks (RNN) a. Implementation of CNN architectures (LeNet and AlexNet) b. Training and performance evaluation on benchmark datasets
5.	Exp 5: Implementation of Recurrent Neural Networks (RNN) a. Implementation of RNN for sequence data b. Implementation of LSTM and GRU networks c. Training and performance evaluation on sequential datasets
6.	Exp 6: Data Transfer Learning a. Importing pre-trained models b. Implementation of pre-trained models with a new learning model c. Performance evaluation and comparison with training from scratch

Evaluation Scheme

Type (Theory + Lab)	Weightage (100)	Exam Max Marks	Remarks	Format
Mid Term	20	50	-	Closed Book, conducted centrally at Univ level
End Term	40	100	-	Closed Book, conducted centrally at Univ level
Lab Assessment	10	-	Best (2 out of 3)	Open Book and/or on LMS or in class
Mini Project	10	-	-	Open Book and/or on LMS take home
Quiz	20	20	Best (1 out of 2)	Closed Book and offline, in class, conducted centrally at School level

1. Evaluation Scheme

ASSESSMENT SCHEDULE PLAN	
ASSESSMENT INSTRUMENTS	**TENTATIVE SCHEDULE**
QUIZ 1	9-11 Feb 2026
QUIZ 2	6-8 April 2026
MIDTERM EXAMINATION	9-14 March 2026

END TERM EXAMINATION	4-19 May 2026